BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

INTERMEDIATE (12TH CLASS) EXAMINATION, 2022

PHYSICS

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Objective Type

Time Allowed: 20 Min. | **Maximum Marks:** 17

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer-book. Cutting or filling two or more circles will result in zero mark in that question.

Note: Objective Type

Q1. The difference of potential energy between two points per unit charge is :

(A) Electrical Potential
(B) Potential difference
(C) Absolute potential
(D) All of these

Q2. The space between the plates of the capacitor is filled by a dielectric of dielectric constant 'k'. The capacitance of the capacitor:

(A) Increased by a factor k
(B) Increased by a factor k2
(C) Decreased by factor k
(D) Remains unchanged

Q3. The current flowing through each resistor of equal resistances in parallel combination is:

(A) Different
(B) Zero
(C) Same
(D) Infinite

Q4. Beam of electron is also called :

(A) X-rays
(B) Alpha rays
(C) Gamma rays
(D) Cathode rays

Q5. Work done on a charge moving in a uniform magnetic field is:

(A) Zero
(B) Positive
(C) Negative
(D) Maximum

Q6. Unit of self inductance is:

(A) Weber
(B) Henry
(C) Tesla
(D) Farad

Q7. The energy stored in the inductor per unit volume is:

(A) B2 / 2 mu0 2
(B) mu0 / 2B
(C) [illegible in source scan]
(D) B2 / 2 mu0

Q8. The mean value of A.C. in one complete cycle is:

(A) 1
(B) Zero
(C) I0
(D) I0 / sqrt(2)

Q9. The most common source of alternating voltage is :

(A) Motor
(B) Cell
(C) Generator
(D) Thermocouple

Q10. The number of crystal systems are :

(A) Three
(B) Five
(C) Fourteen
(D) Seven

Q11. The Boolean expression of NAND gale is :

(A) X = A.B
(B) X = A bar
(C) X = bar over (A.B)
(D) X = bar over (A + B)

Q12. The devices which are used to convert various physical quantities into electrical voltages are called:

(A) Filters
(B) Sensor
(C) Rectifiers
(D) Amplifiers

Q13. Light emitting diodes (LEDs) are made from semi-conductors :

(A) Silicon
(B) Germanium
(C) Gallium arsenide
(D) Carbon

Q14. Compton effect is associated with :

(A) Gamma rays
(B) Beta rays
(C) X-rays
(D) Positive rays

Q15. In electronic transition, atom cannot emit:

(A) Infrared radiations
(B) Visible radiations
(C) Gamma radiations
(D) Ultraviolet radiations

Q16. Energy released by conversion of 1 amu of mass is :

(A) 1.6 x 10-19 ev.
(B) 1.6 x 10-19 Mev
(C) 200 Mev
(D) 931 Mev

Q17. Alpha particle carries a charge of :

(A) +2e
(B) - 2e
(C) +e [watermarked, uncertain]
(D) Zero

Essay Type

Time Allowed: 2.40 hrs. | **Maximum Marks:** 68

Note: SECTION - I

Q2. (Marks 16) Write short answers to any EIGHT (8) questions:

(i) Write down any two properties of electric field lines.

(ii) State Coulomb's law and Gauss's law.

(iii) Suppose that you follow an electric field line due to a positive point charge. Do electric field and the potential increase or decrease?

(iv) Do electrons tend to go to region of high potential or of low potential?

(v) Define stable or dead beat galvanometer.

(vi) Differentiate between magnetic flux and magnetic flux density. Also write units of both.

(vii) Two charged particles are projected into a region where there is a magnetic field perpendicular to their velocities. If the charges are deflected in opposite directions, what can you say about them?

(viii) How can a current loop be used to determine the presence of a magnetic field in a given region of space?

(ix) Differentiate between hadrons and leptons. Also give examples of each.

(x) Enlist the basic forces of nature.

(xi) What factors make fusion reaction difficult to achieve?

(xii) A particle which produces more ionization is less penetrating. Why?

Q3. (Marks 16) Write short answers to any EIGHT (8) questions:

(i) What are the difficulties in testing whether the filament of lighted bulb obeys Ohm's law?

(ii) Define temperature co-efficient of resistance and write its formula.

(iii) Prove that : Volt x Ampere = Watt.

(iv) What is meant by A.M. and F.M.?

(v) What is the main advantage of three phase A.C. supply?

(vi) What is difference between A.C. circuit and D.C. circuit?

(vii) Draw a stress-strain curve for a ductile material and then define the terms: (i) Elastic limit (ii) Ultimate tensile stress.

(viii) What are the two main differences between conductors and semi-conductors?

(ix) Describe energy band picture of insulators.

(x) Why charge carriers are not present in the depletion region?

(xi) Give four applications of a photodiode.

(xii) How is p-n junction formed?

Q4. (Marks 12) Write short answers to any SIX (6) questions:

(i) State Faraday's law of electromagnetic induction.

(ii) What is back emf effect in motor?

(iii) Show that epsilon and delta phi / delta t have the same units.

(iv) Can an electric motor be used to drive an electric generator with the output from generator being used to operate the motor?

(v) Explain uncertainty principle.

(vi) Write four uses of laser in medicine and industry.

(vii) What do you mean when we say that the atom is excited?

(viii) What is the advantage of NAVSTAR navigation system?

(ix) What happens to total radiation from a black body, if its absolute temperature is doubled?

> Note: SECTION-II

> Note: Note: Attempt any THREE questions.

Q5. (a) What is Wheatstone Bridge? How Wheatstone Bridge can be used to determine an unknown resistance? 1,4 [sic]

(b) A particle having charge of 20 electrons on it fails [sic, print for 'falls'] through a potential difference of 100 volts. Calculate the energy acquired by it in electron volts (ev). 3

Q6. (a) How can you determine e/m of an electron? Explain how the path of electron beam is made visible? 5

(b) An emf of 5.6 V is induced in a coil while the current in a nearby coil is decreased from 100 A to 20 A in 0.02s. What is mutual induction of two coils? If secondary coil has 200 turns, find change in flux during this interval. 3

Q7. (a) Discuss RLC series circuit. Derive the formula for resonance frequency. Also properties of this circuit. 5

(b) The current flowing into the base is 100 microA. Find its collector current IC, its emitter current IE and IC / IE if 'beta' current gain is 100. 3

Q8. (a) What is energy band theory? Explain the difference amongst electrical behaviour of conductors, insulators and semi-conductors in terms of energy band theory. 5

(b) What is the de-Broglie wavelength of an electron whose kinetic energy is 120eV? 3

Q9. (a) Derive an expression for the energy of electron revolving in nth orbit of hydrogen atom. 5

(b) A sheet of lead 5 mm thick reduces the intensity of beam of gamma-rays by a factor 0.4. Find half value thickness of lead sheet which will reduce the intensity to half of its initial value. 3

GROUP – II

Objective Type

Time Allowed: 20 Min. | **Maximum Marks:** 17

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer-book. Cutting or filling two or more circles will result in zero mark in that question.

Note: Objective Type

Q1. Electric flux through a closed surface enclosing a charge depends on :

(A) Medium
(B) Size
(C) Shape
(D) Location of charge

Q2. Unit of electric intensity is same as:

(A) Force
(B) Potential gradient
(C) Viscosity
(D) Magnetic field

Q3. Thermo-couples produce electric energy by:

(A) Heat
(B) Chemical energy
(C) Sunlight
(D) Mechanical energy

Q4. Work done by a magnetic force of 5 N when a q charge is displaced 2 m is:

(A) Non-zero
(B) Zero
(C) 10 J
(D) 5 J

Q5. E / B has the unit of:

(A) meter
(B) ms-1
(C) ms-2
(D) s-2

Q6. A metal meter rod is moving at the speed of 0.5 ms-1 in the direction parallel to a 0.5T magnetic field, emf will be:

(A) 0.25 V
(B) 0.5 V
(C) Zero
(D) 0.125 V

Q7. Lenz's law is also a statement of law of conservation of:

(A) Linear momentum
(B) Angular momentum
(C) Energy
(D) Charge

Q8. If the frequency of A.C. is 40 Hz then current passing through filament bulb get brilliance:

(A) 100 times
(B) 80 times
(C) 40 times
(D) 50 times

Q9. If V0 is peak value of A.C. voltage then mean square value of voltage is:

(A) V0 / sqrt(2)
(B) V0 2
(C) 1/2 V0 2
(D) V

Q10. In cubical crystal, all the sides meet at:

(A) Acute angle
(B) Abtuse angle [sic, print for 'Obtuse angle']
(C) Right angle
(D) 45 degrees

Q11. When PN junction is conducting then its resistance is of the order of :

(A) Mega Ohm
(B) Kilo Ohm
(C) 100 Ohm
(D) Few Ohms

Q12. [battery symbol with diagonal arrow (variable), partially watermarked] is symbol of:

(A) High tension battery
(B) Low tension battery
(C) Variable voltage battery
(D) Zero resistance battery

Q13. The observations on objects moving very fast, approaching the speed of light, arc [sic, print for 'are'] well explained by :

(A) Quantum theory
(B) Newton's law
(C) Special theory of relativity
(D) Kepler's law

Q14. A block body [sic, print for 'black body'] is both an ideal absorber and an ideal :

(A) Reflector
(B) Radiator
(C) Conductor
(D) Insulator

Q15. Plank's [sic, print for "Planck's"] constant has the unit of ::

(A) Linear momentum
(B) Angular momentum
(C) Torque
(D) Force

Q16. Two quark combination forms :

(A) Mesons
(B) Baryons
(C) Leptons
(D) No Composite particle

Q17. Energy given out per nucleon per fission of heavy element like uranium is ::

(A) 200 MeV
(B) 208 MeV
(C) 5 MeV
(D) 0.9 MeV

Essay Type

Time Allowed: 2.40 hrs. | **Maximum Marks:** 68

> Note: SECTION - I

Q2. (Marks 16) Write short answers to any EIGHT (8) questions:

(i) What is meant by electric polarization?

(ii) Prove that electric intensity inside a hollow charged sphere is zero.

(iii) Electric lines of force never cross each other, why?

(iv) How can you identify that which plate of the capacitor is positively charged? Explain it.

(v) Write down any four uses of CRO.

(vi) What is Lorentz force? Write down its formula.

(vii) Why does the picture on the screen is distorted when a magnet is brought near its screen?

(viii) How a galvanometer can be made sensitive?

(ix) What is the binding energy? Write down the name of element which has highest value.

(x) Heavy nuclei are unstable why?

(xi) What do you mean by dead time in Geiger-Muller Counter?

(xii) What factors make a fusion reaction difficult to achieve?

Q3. (Marks 16) Write short answers to any EIGHT (8) questions:

(i) Write down four sources of current.

(ii) Do bends in a wire affect its electrical resistance? Explain.

(iii) Is the filament resistance lower or higher in a 500 w, 220 volt bulb than in a 100 w, 220 volt bulb?

(iv) Define the terms peak value and peak to peak value.

(v) Discuss two uses of three phase A.C. supply.

(vi) How the reception of a particular radio station is selected on your radio set?

(vii) Explain ductile substances and brittle substances.

(viii) What is meant by hysteresis loss?

(ix) Show that units of modulus of elasticity and stress are the same.

(x) Why charge carriers are not present in the depletion region?

(xi) What is the principle of virtual ground?

(xii) Calculate the gain of a non-inverting amplifier. When R1 = infinity and R2 = zero.

Q4. (Marks 12) Write short answers to any SIX (6) questions:

(i) Can a step up transformer increase the power level? Explain.

(ii) How would you position a flat loop of a wire in changing magnetic field, so that there is no emf induced in the loop?

(iii) Write down the factors upon which the mutual inductance depend.

(iv) Distinguish between A.C. generator and transformer.

(v) Will higher frequency light eject greater number of electrons than low frequency light?

(vi) When does light behave as a wave? When does it behave as particle?

(vii) State Stefen's [sic, print for "Stefan's"] Boltzman law. Also write the value of Stefen's constant?

(viii) Find the shortest wavelength of radiation in the Balmer series.

(ix) What do we mean when we say that the atom is excited?

> Note: SECTION - II

> Note: Note: Attempt any THREE questions.

Q5. (a) State and explain Gauss's law, also calculate the electric intensity due to an infinite sheet of charge. 5

(b) The resistance of an iron wire at 0C is 1 x 10^4 Ohm. What is the resistance at 500C, if the temperature coefficient of resistance of iron is 5.2 x 10^-3 K^-1? 3

Q6. (a) State Ampere's law and apply it to find magnetic field due to a current carrying solenoid. 5

(b) A solenoid has 250 turns and its self inductance is 2 mH. What is the flux through each turn when current is 2 A? What is the induced emf when current changes at 20 As^-1? 3

Q7. (a) Write a note on transistor as an amplifier. 5

(b) A circuit has an inductance of 1/pi H and resistance of 2000 Ohm. A 50 Hz A.C. is supplied to it. Calculate the reactance and impedance offered by the circuit. 3

Q8. (a) Define photoelectric effect. Give its explanation on the basis of Quantum theory. 5

(b) A wire 2.50 m long and cross-section area 10^-5 m^2 is stretched 1.5 mm by a force of 100 N in the elastic region. Calculate Young's modulus. 3

Q9. (a) What is laser? Write down its properties and also explain laser action in detail. 5

(b) Find the mass defect and binding energy of the deuteron nucleus. The experimental mass of deuteron is 3.3435 x 10^-27 kg. 3